

Midterm exam, June 11th 2026

KSE, Algebraic topology, Summer 2026

Length: 2 hours

Credits: 30 points

Exercise 1. Null-homotopy and extension to the cone

Let X be a space, recall that the cone $C(X)$ is defined as $X \times [0, 1]/X \times 0$, and there exists a canonical inclusion $X \cong X \times 1 \subset C(X)$.

For a map $f : X \rightarrow Y$, we say f is *null-homotopic* if f is homotopic to the constant map. Show that $f : X \rightarrow Y$ is null-homotopic if and only if f can be extended to a map $\hat{f} : C(X) \rightarrow Y$ such that $\hat{f}|_{X \times 1} = f$.

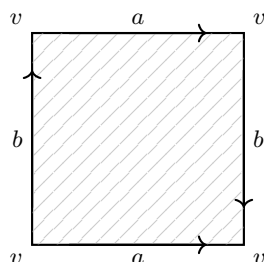
Exercise 2. About the k -punctured plane

Let $X = \mathbb{R}^2 \setminus \{p_1, \dots, p_k\}$ for distinct points $\{p_1, \dots, p_k\} \subset \mathbb{R}^2$.

Compute the homology groups $H_q(X)$ for all $q \geq 0$. What can you say about the homotopy type of X ? You can draw some pictures to illustrate homotopies if you are not able to write down an explicit homotopy.

Exercise 3. Homology of the Klein bottle

Let K be the Klein bottle defined by the following (where the square is filled by a 2-cell):



Compute the homology groups $H_q(K)$ for all $q \geq 0$, using your favorite way.

Exercise 4. About the Möbius band

Let $M = [0, 1] \times [-1, 1]/((0, t) \sim (1, -t))$ be the Möbius band, and let $S = [0, 1] \times \{0\}/\sim \subset M$ be its core circle.

- (a) Show that $S \cong S^1$ is a strong deformation retract of M .
- (b) Compute the homology groups $H_q(M)$ for all $q \geq 0$.
- (c) Let $\partial M = \{[u, \pm 1] \in M \mid u \in [0, 1]\}$ be the boundary and $j : \partial M \hookrightarrow M$ be the inclusion of the boundary. Show that $\partial M \cong S^1$ and compute the induced map

$$j_1 : H_1(\partial M) \longrightarrow H_1(M).$$

Here, compute means that you should write those groups as some precise abelian groups, and write down the map explicitly.